NBA Post-Rookie Forecasting and Extension Valuation

STAT 946 Case Study 3

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Objective

Build a multi-module player future analysis system that uses a rookie's first 4 NBA seasons to support post-rookie extension decisions.

Instead of producing only one forecast, we build a compact decision system that evaluates:

- future performance
- future durability
- contract value
- indicator of potential sleeper / neutral / bust
- front-office action

Core outputs for decision recommendation include but not limited to:

- Post-rookie performance forecast
- Post-rookie contract value as salary-cap %
- Player Archetype & Comparable Profile

Target Audience and Our Roles

Primary audience

- NBA front offices
- Basketball operations groups
- Contract strategy teams

Secondary audience

Players and agents evaluating contract asks and market position

Who we are

Senior Data Scientists in the Basketball Analytics group of an NBA franchise

Final deliverables

Our approach: a multi-module, role-aware forecasting system

Traditional approach: aggregate box-score stats + linear / logistic regression, common in basketball analytics because box-score data are the most available and easy to interpret.

- **Main limitation:** averages flatten the player's path and can miss important signals in the development trajectory, usage, and role evolution.

We present the project as **3 main final deliverables**

Deliverable 1 — Performance Forecast

What we investigate

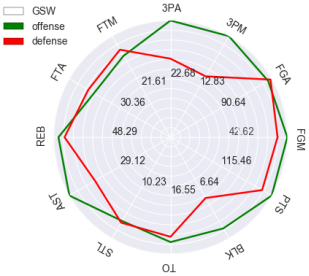
- Years 5–7 basketball performance
- Sleeper / Neutral / Bust outcome
- uncertainty around the forecast

Method

- baseline logistic regression / classification models first
- LSTM hybrid sequence model on Seasons 1-4 game-by-game logs
- confidence band from repeated splits / bootstrap / ensemble-style uncertainty

Final output

- predicted Years 5-7 production
- sleeper / neutral / bust probabilities
- floor / base / ceiling forecast
- skillset radar chart (showcase example)



Source: Jacobs (2016)

Deliverable 2 — Player Archetype & Comparable Profile

What we investigate

- current player archetype
- identity drift / role evolution
- realistic comparable players
- historical role model
- similarity score and comp-group outcome context

Method

- **PCA + K-means** for early-career archetypes
- nearest neighbors in PCA / archetype space for realistic comps
- post-1990 Hall of Fame library for historical role models
- **ShotChartDetail core pipeline**
  - pull Seasons 1-4 shot spatialtemporal data (LOC_X, LOC_Y, shot type, clock, period, result)
  - bin shots into player-season spatial tensors
  - train a CNN autoencoder
  - export shot-style embeddings
  - use embeddings for comps and shot-style drift

Final output

- archetype label + archetype match score
- realistic comps + historical role models if justified
- similarity score
- median later-career outcome of realistic comp group
- role / shot-style evolution summary

Deliverable 3 — Salary Decision Support

What we investigate

- Year-5 contract value as salary-cap %
- durability / missed-time risk
- final extension stance

Method

- merge player salary history with salary-cap history
- baseline salary regression, with optional LSTM salary head if coverage is stable
- durability risk model from games, minutes, rest gaps, and participation stability
- rules-based recommendation quadrant combining:
  - forecast
  - salary value
  - durability risk
  - comps
  - uncertainty

Final output

- Year-5 salary-cap % forecast
- durability / risk band
- recommendation:
  - Extend Aggressively
  - Extend at Disciplined Price
  - Wait and Reassess
  - Avoid Overcommitting

Why a multi-module system?

Data Limits

- NBA downstream targets are player-level, not million-row data
- NBA's official player-tracking data became league-wide in the 2013–14 season, and current NBA tracking pages still note that tracking is not available for all games
- not all players have full Years 5–7 outcomes
- current files are boxscore-based, frame-by-frame tracking spatiotemporal data is mostly locked behind NBA/Second Spectrum

So instead of forcing all sophistication into one unified model, we aim for more coverage, compactness, and decision value across several modules:

- coverage across multiple decision angles
- compactness in one integrated player dossier
- honesty about uncertainty and sample attrition
- business value for post-rookie contract decisions

Background: The Business Story

Why Use Seasons 1–4

- The first four seasons match the rookie-contract window, when teams must judge long-term value
- Rookie extension decisions are high-stakes and made before full career outcomes are visible
- Season 1–4 data contain early signals on usage, efficiency, durability, and development path

We frame this as an **RFA war room** problem: pay now, proceed carefully, or avoid overcommitting

Forecast Window

- Years 5–7 are a natural post-rookie window to evaluate whether early signals translate into sustained value

Workflow

1. **Cohort + targets** Years 5–7 performance, Year-5 cap %, Sleeper/Neutral/Bust
2. **Feature engineering** player-level table + game-sequence table + durability features
3. **Archetype learning** PCA + K-means ($k = 5$) for early-career player types
4. **Shot-style module** ShotChartDetail -> spatial tensors -> CNN autoencoder -> shot embeddings
5. **Archetype & comps** archetype, role evolution, realistic comps, HOF ceiling comps
6. **Performance forecast** baseline models + K-fold LSTM hybrid + uncertainty band
7. **Salary decision support** salary-cap % forecast + durability risk + recommendation quadrant
8. **Final deliverables** Performance Forecast / Archetype & Comparable Profile / Salary Decision Support

Data Resources and Scraping

Cohort: 1247 drafted players from 1999-2019 (from *nba_api*)

Rookie-contract logs: Seasons 1-4 game-by-game NBA boxscore data (179819 games) to build player-level features and sequence data (from *nba_api*)

Later-career total (targets): realized player-season outcomes used to build post-rookie years performance labels and contract value labels (from *nba_api* and *basketball-reference.com*)

Contract data: Scrape per player per year salary data from Hoopshype (1990 - 2019) + Historical NBA salary cap table to create salary-cap % target (from *https://www.hoopshype.com* and *https://www.basketball-reference.com/contracts/salary-cap-history.html*)

Historical comps: scrape post 1990 "Hall of Fame" players career trajectories (Season 1-4 game-by-game box scores + later-career outcomes) (from *https://www.nba.com/news/basketball-hall-of-fame-all-time* and *nba_api*)

ShotChartDetail: nba_api - scrape per-shot spatiotemporal records with LOC_X and LOC_Y, plus shooter, period, clock, shot type, result, and distance. (requires roughly 4,000 API calls ~1.4 Million raw spatial events, not yet integrated)

PlayByPlayV3: nba_api - event-level spatiotemporal timeline data with clock, period, descriptions, and legacy shot/event coordinates xLegacy / yLegacy (optional, not yet integrated)

NCAA Game logs: scrape players' NCAA game logs for potential early-career signals (optional, not yet integrated)

Injury data: scrape players' injury history for durability features and potential injury-risk signals (optional, not yet integrated)

PCA: turning many features into a compact player space

Input: Seasons 1–4 flattened player feature table

Features used: four-year production, efficiency, workload, slopes, and season-block summaries

Preprocessing: winsorize outliers, median impute missing values, standardize features

Method: PCA, keeping enough components to explain 85% of variance

Purpose: reduce noise and collinearity before clustering

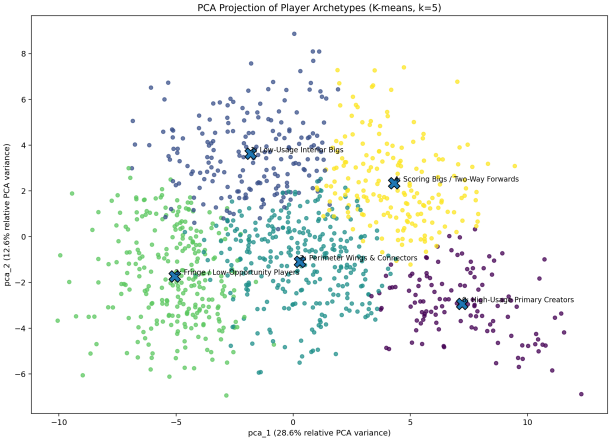
Output: low-dimensional PCA scores for each player

Use in project: direct input to K-means archetype clustering and comparable-player analysis

PCA + K-means Archetype Model

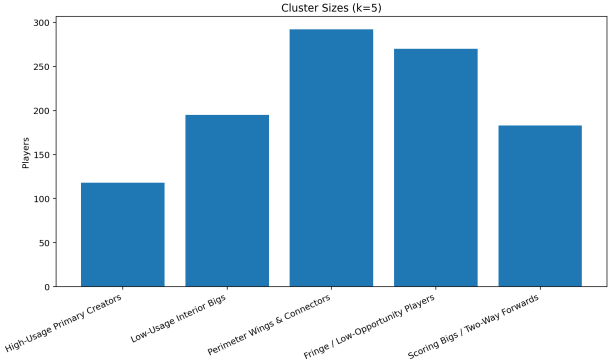
- Clustered **1,058 players** using **24 PCA components** and **$k = 5$** clusters
- Silhouette score in PCA space: **0.107**
- Archetypes found:
  - High-Usage Primary Creators (118): High assists, strong scoring, high usage, meaningful minutes
  - Low-Usage Interior Bigs (195): High rebounds and blocks, low 3PA tendency, lower usage
  - Perimeter Wings & Connectors (292): Wing-like shooting profile, moderate playmaking, balanced usage
  - Fringe / Low-Opportunity Players (270): Lowest minutes and weakest opportunity profile
  - Scoring Bigs / Two-Way Forwards (183): Strong scoring, rebounding / block profile, more frontcourt production

PCA + K-means Results Snapshot



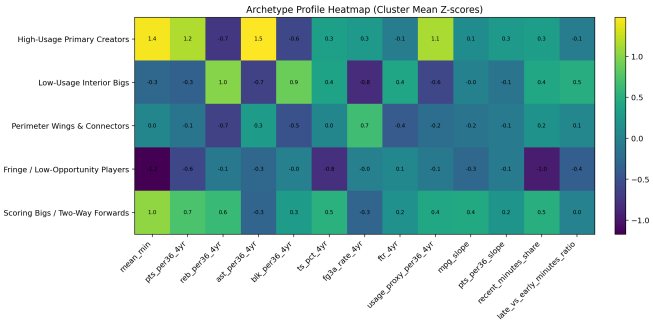
- PCA gives a compact player map
- K-means learns **5 role-based archetypes**
- Archetypes become model inputs:
  - one-hot clusters
  - centroid distances
- Better peer-group comparison
- Better stakeholder interpretation

Cluster Size and Sample Balance



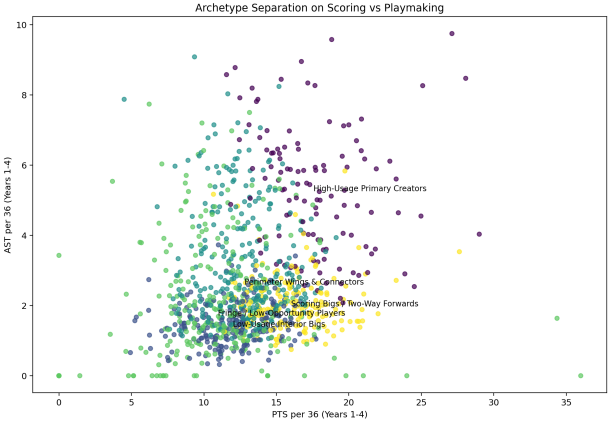
- No archetype is tiny or degenerate
- Largest groups are "Fringe / Low-Opportunity" and "Perimeter Wings & Connectors"
- Smallest group is "High-Usage Primary Creators", which is expected because true initiators are rarer
- This gives us usable coverage across all 5 archetypes for downstream modeling

How the 5 Archetypes Differ



- Heatmap shows **cluster mean z-scores**
- Compare archetypes on the same standardized scale
- Main dimensions include:
  - minutes / opportunity
  - scoring
  - playmaking
  - rebounding / rim protection
  - shooting profile
  - usage and development path

Intuitive Basketball View: Scoring vs Playmaking



- Plots archetypes using **PTS per 36** and **AST per 36**
- Gives a simple basketball interpretation of the clusters
- **Primary Creators** separate through higher assist load
- **Scoring Bigs / Two-Way Forwards** separate through scoring without creator-level passing
- **Fringe players** stay low on both dimensions
- Helps explain why pooled league-wide comparisons can be misleading

Why the Archetypes Matter Downstream

Archetypes are **not just side visuals**

They become model inputs through:

- **cluster one-hot indicators**
- **distance-to-centroid variables**
- **own-cluster distance** as a prototype-fit / ambiguity measure

This helps us:

- compare players within the **right peer group**
- improve baseline models with role-aware predictors
- concatenate archetype information into the **LSTM hybrid model**
- support comparable-player search and player dossier storytelling

Game Data Available - Shooting Statistics

- FGM (Field Goals Made): Total number of baskets made (includes both 2-pointers and 3-pointers, but not free throws).
- FGA (Field Goals Attempted): Total number of shots taken.
- FG_PCT (Field Goal Percentage): The percentage of shots made (FGM / FGA).
- FG3M (3-Point Field Goals Made): Number of 3-point shots made.
- FG3A (3-Point Field Goals Attempted): Number of 3-point shots taken.
- FG3_PCT (3-Point Percentage): The percentage of 3-point shots made (FG3M / FG3A).
- FTM (Free Throws Made): Number of successful free throws.
- FTA (Free Throws Attempted): Number of free throws taken.
- FT_PCT (Free Throw Percentage): The percentage of free throws made (FTM / FTA).

Game Data Available - Rebounding & Playmaking

- OREB (Offensive Rebounds): Rebounds grabbed while the player's team is on offense (giving them another chance to score).
- DREB (Defensive Rebounds): Rebounds grabbed while the player's team is on defense.
- REB (Total Rebounds): The sum of Offensive and Defensive rebounds (OREB + DREB).
- AST (Assists): Passes that lead directly to a teammate scoring a basket.

Game Data Available - Defense & Mistakes

- STL (Steals): Number of times the player legally takes the ball away from an opponent.
- BLK (Blocks): Number of times the player successfully deflects an opponent's shot attempt.
- TOV (Turnovers): Number of times the player loses possession of the ball to the other team (via a bad pass, traveling, getting stolen from, etc.).
- PF (Personal Fouls): Number of fouls committed by the player in that game.

Game Data Available - Overall Impact

- PTS (Points): Total points scored by the player in that game.
- PLUS_MINUS (+/-): A metric showing the team's point differential while that specific player was on the court. A "+8" means the team scored 8 more points than the opponent while Zion was playing. A "-21" means they were outscored by 21 points while he was on the floor.

Player Performance Analysis

- We want to calculate the player performance for each game & for each season/year.
- Several Options
- Game Score: created by John Hollinger, this is essentially a "single-game PER" (Player Efficiency Rating). It gives a rough, all-in-one number indicating how productive a player was in a specific game. A score of 10 is average, and 40 is an all-time great performance.

$$GmSc = PTS + 0.4 * FGM - 0.7 * FGA - 0.4 * (FTA - FTM) + 0.7 * OREB + 0.3 * DREB + STL + 0.7 * AST + 0.7 * BLK - 0.4 * PF - TOV$$

Player Performance Analysis

- Value Point System: based on a formula that assesses performance based upon several offensive and defensive stats. The higher the number, the better. A VPS of 1 is about average.

$$VPS = \frac{PTS+REB+2*AST+2*(STL+BLK)}{2*(FGA-FGM)+(FTA-FTM)+2*PF+2*TOV}$$

- True Shooting Percentage: measures a player's efficiency at shooting the ball across all types of shots.

$$TS\% = \frac{PTS}{2*(FGA+0.44*FTA)}$$

Multinomial logistic regression

- We use multinomial logistic regression to predict a player's final career outcome class from their first four years of NBA data.
- The three classes are Sleeper, Neutral, and Bust.
- The goal is to see whether early-career opportunity, efficiency, production, trend, and stability can help explain which direction a player's later career will go.

What we need to do

1. Build the outcome score as Y

- Use career total performance data
- Turn the scores into final labels: Sleeper / Neutral / Bust

2. Decide the predictor variables as X

- Use information from the player's first 4 seasons
- Include opportunity, efficiency, production, trend, stability, and cluster features

3. Fit multinomial logistic regression

- Predict the three outcome classes from early-career features

4. Evaluate model performance

- Check accuracy, macro-F1, and confusion matrix
- See how well the model separates Sleeper, Neutral, and Bust

How We Found the Outcome Scores

- Summarize long-term career performance base on player performance analysis
- Apply PCA to create realized outcome scores
- Compare actual score with expected score, we used the ridge regression
- Use residuals to measure over- or under-performance
- Classify each player as Sleeper / Neutral / Bust
- Use majority vote to get the final class label
- "Sleeper": 0, "Neutral": 1, "Bust": 2

How We Built the Multinomial Logistic Regression

- Use the final label as the target variable, with three classes: Sleeper, Neutral, and Bust.
- Use selected early-career features as predictors, including opportunity, efficiency, per-36 production, trend, stability, and cluster information.
- Treat the cluster variable as categorical using one-hot encoding, and standardize the numeric predictors before modeling.
- Fit a multinomial logistic regression model and evaluate it with metrics such as accuracy, macro-F1, and the confusion matrix.

Result from Multinomial Logistic Regression

```
Multinomial Logistic Regression Results

Accuracy   : 0.5077
Macro-F1   : 0.4727

Classification Report:
              precision    recall  f1-score   support

           0     0.4516    0.5185    0.4828        27
           1     0.4400    0.2558    0.3235        43
           2     0.5541    0.6833    0.6119        60

    accuracy                         0.5077       130
   macro avg     0.4819    0.4859    0.4727       130
weighted avg     0.4951    0.5077    0.4897       130
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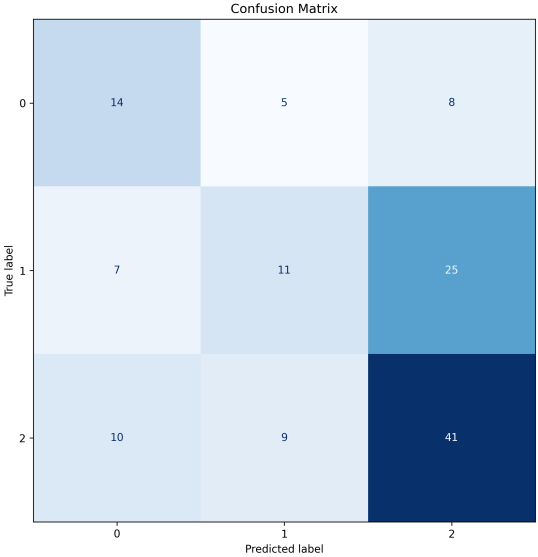
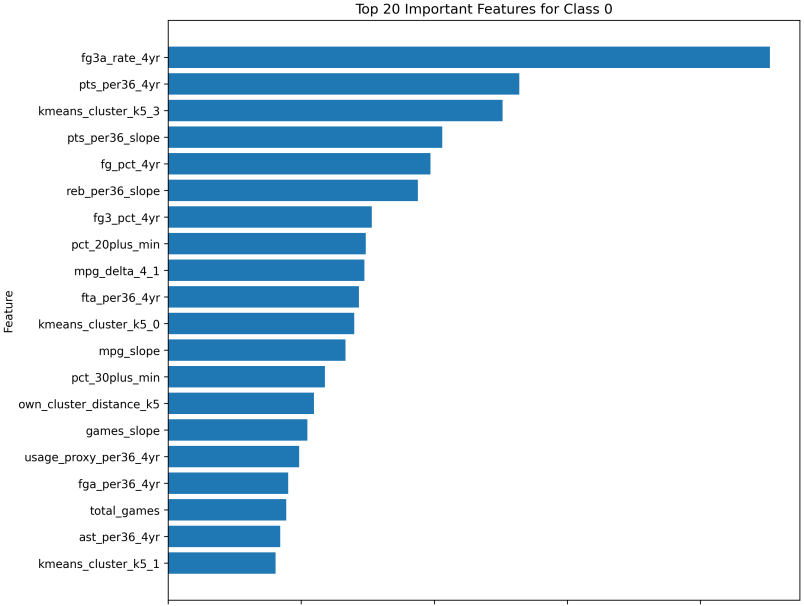
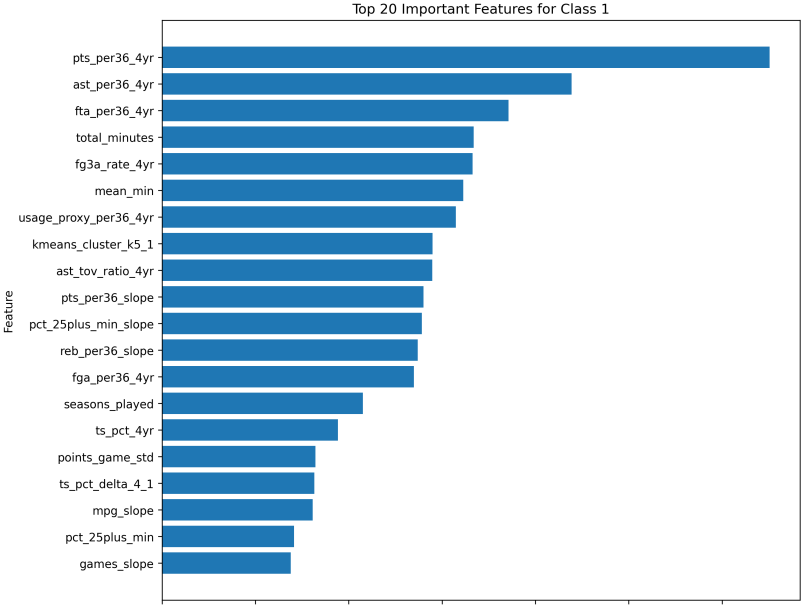


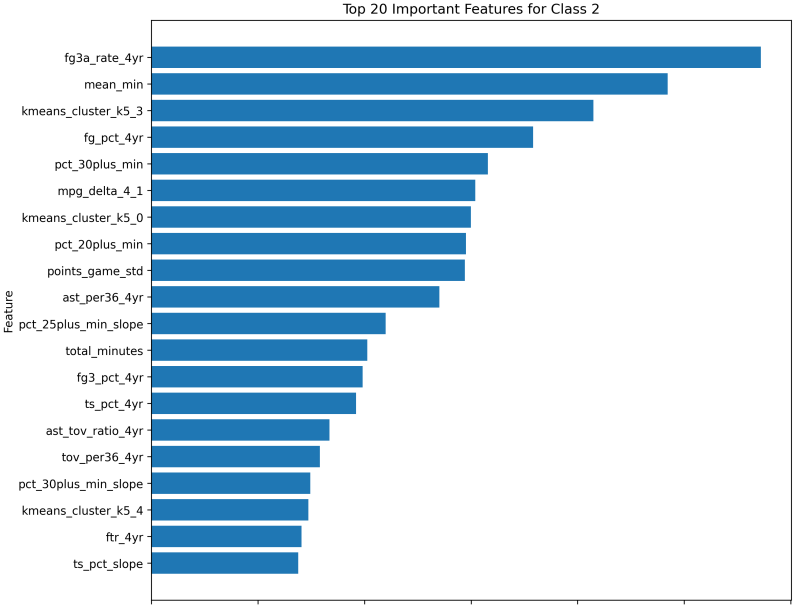
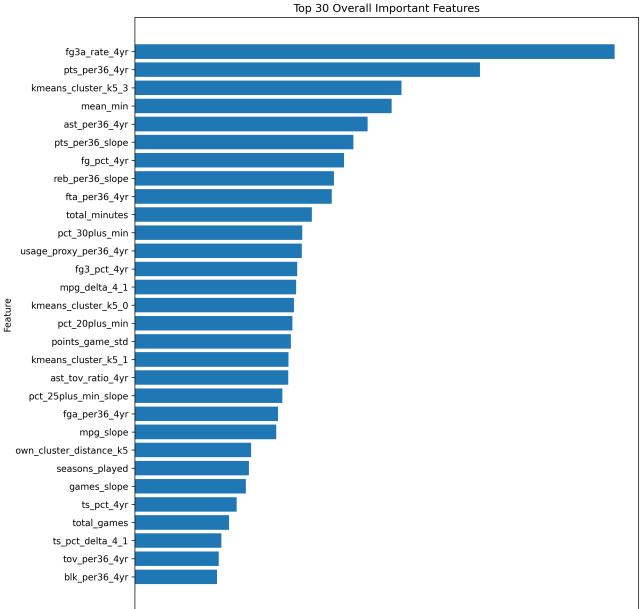
Feature importance for class 0



Feature importance for class 1



Feature importance for class 2



Feature importance Overall